



CERTIFICATE

This is to certify that Ms Kathrin Markworth, born on December 20, 1982,
has successfully completed our correspondence course



Certified Java Programmer (ILS)

(„Geprüfte Java-Programmiererin (ILS)“)

approved by the state authority for distance learning (course no: 7179107) and taking place
between June 13, 2022 to February 05, 2023.

The curriculum has a total of 580 time hours including examination achievements.

Ms Markworth has submitted written assignments in all subject areas and a final test paper.

- introduction to programming structures
- using the array class
- developing object-oriented software
- creating programs with graphical user interface
- troubleshooting and exception handling
- using XML-, binary- and text files
- database programming
- building and distributing applets and hybrid applications
- programming projects: memory game, word processing, drawing program, address management

The proctored examinations in the seminars were passed with a grade point average of 1,0*.

Ms Markworth has been awarded an overall grade of 1,0.*

*(*The German grading scale uses the numbers 1 to 6, with 1 representing the highest, 6 the lowest grade.
The use of a decimal point gives a more exact grading.)*

Hamburg, February 09, 2023

Director of Learning

Dipl.-Volksw. Jens Greefe

Director of Studies

Dipl.-Kfm. Werner Müller